

Supporting Information:

Supporting Information for VIAMD–VeloxChem

Integration for Interactive Quantum Chemical

Visualization and Analysis

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Datasets and Feature Matrix

The examples used in this article are based on the VeloxChem input/output file collection available from the VeloxChem documentation at <https://veloxchem.org/docs/input-files/>.

This collection provides paired input scripts, text input files, log files, and HDF5 output files for representative VeloxChem calculations. The HDF5 files are regenerated regularly from the VeloxChem master branch, and the VIAMD developers use this collection as a practical test set for checking the VeloxChem interface and the visualization features described in the main article.

Table S1 follows the organization of the VeloxChem input/output page and summarizes the VIAMD windows and feature families associated with each example. In the table, the Common column denotes the shared VIAMD starting point for VeloxChem HDF5 files: direct file loading; molecular-structure inspection; atom selection and highlighting through interactive tables; calculation metadata and convergence review in the Summary window; standard molecular representations in the spatial view; molecular-orbital, density, and density-difference representations when the required electronic-structure data are present; Orbital Grid comparison when orbital data are present; molecular vectors or axes when available; density topology and critical-point inspection when the density can be reconstructed; and saving or restoring the analysis state through VIAMD workspaces. The remaining columns mark additional interactive workflows available for specific examples. Vibrational spectra are included under the same Response workflow as electronic and other response-property spectra. The final column is restricted to atom-centered property datasets, such as ESP, RESP, Boltzmann-weighted RESP, and LoProp quantities.^{S1-S3}

Table S1: Feature matrix for VeloxChem input/output HDF5 files used to test and illustrate the VIAMD VeloxChem interface. The entries follow the order of the VeloxChem input/output documentation page. A bullet in the Common column denotes the shared VIAMD starting point: HDF5 loading, Summary-window metadata, molecular/electronic representations, Orbital Grid access when orbital data are present, molecular vectors or axes when present, density topology when density data allow it, and workspace saving. Optimisation marks geometry optimization, constrained-coordinate, scan, transition-state, and IRC workflows. Response covers electronic, vibrational, CPP, X-ray, and multiphoton response workflows. Transition marks attachment/detachment and charge-transfer analysis. Atomic props. marks atom-centered property datasets such as ESP, RESP, Boltzmann-weighted RESP, and LoProp. Feature availability should be verified against the monthly regenerated HDF5 files before final publication.

Example	HDF5 file	<i>Common</i>	<i>Optimisation</i>	<i>Response</i>	<i>Transition</i>	<i>Atomic props.</i>
SCF optimization	biphenyl-scf.h5	•	—	—	—	—
ROHF optimization	tempo-roscf.h5	•	—	—	—	—
UHF optimization	trityl-uscfc.h5	•	—	—	—	—
Static electric field	pna-field.h5	•	—	—	—	—
CPCM environment	eth-cpcm.h5	•	—	—	—	—
SMD environment	eth-smd.h5	•	—	—	—	—
PE/NPE environment	alpha-helix-acetone-water-pe.h5	•	—	—	—	—
S0 optimization	bithio-S0-opt.h5	•	•	—	—	—
Frozen coordinate	bithio-freeze.h5	•	•	—	—	—
Coordinate scan	bithio-scan.h5	•	•	—	—	—
TS optimization	Sn2-ts.h5	•	•	—	—	—
IRC	Sn2-irc.h5	•	•	—	—	—
S1 optimization	bithio-S1-opt.h5	•	•	•	—	—

Example	HDF5 file	<i>Common</i>	<i>Optimisation</i>	<i>Response</i>	<i>Transition</i>	<i>Atomic props.</i>
TDDFT UV/Vis	tq-uv-vis.h5	•	—	•	•	—
CPP UV/Vis	tq-cpp.h5	•	—	•	—	—
Ground-state dipole	pna-GS-dipole.h5	•	—	—	—	—
Excited-state dipole	pna-ES-dipole.h5	•	—	•	—	—
Nonresonant polarizability	meth-nonres.h5	•	—	•	—	—
Resonant polarizability	eth-res.out	—	—	—	—	—
TDDFT optical activity	alanine-ecd.h5	•	—	•	—	—
CPP optical activity	alanine-cpp.h5	•	—	•	—	—
IR spectroscopy	acro-ir.h5	•	—	•	—	—
Raman spectroscopy	acro-raman.h5	•	—	•	—	—
Resonance Raman	acro-rr.h5	•	—	•	—	—
C6 weak interactions	meth-c6.h5	•	—	—	—	—
ESP	h2o-esp.h5	•	—	—	—	•
RESP	h2o-resp.h5	•	—	—	—	•
Boltzmann-weighted RESP	pro-resp-bw.h5	•	—	—	—	•
LoProp	h2o-loprop.h5	•	—	—	—	•
TPA	pna-tpa.h5	•	—	•	—	—
TPA cross section	pna-tpacs.h5	•	—	•	—	—
XPS	esca-xps.h5	•	—	•	—	—
XAS (CVS)	xas-cvs.h5	•	—	•	—	—

Example	HDF5 file	<i>Common</i>	<i>Optimisation</i>	<i>Response</i>	<i>Transition</i>	<i>Atomic props.</i>
XAS (CPP)	<code>esca-nexafs.h5</code>	•	–	•	–	–
RIXS	<code>esca-rixs.h5</code>	•	–	•	–	–
General response functions	<code>h2o2-lrf.out,</code> <code>h2o2-crf.out</code>	–	–	–	–	–

Together, the documentation collection and the feature matrix define a reproducible route from VeloxChem calculations to VIAMD analysis: the main article highlights representative examples for the main workflows, while the full set remains available here as a practical regression and tutorial resource as the interface evolves.

References

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